A PROJECT REPORT

ON

"IMAGE COMPRESSION MATRIX RANK REDUCTION"

Submitted
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DECLARATION

We, **J SPANDANA (26RH1A05R9), J VIJETHA (26RH1A05T0), J SUCHITHRA(26RH1A05T1),** hereby declare that the project entitled **"IMAGE COMPRESSION MATRIX RANK REDUCTION"** submitted by us is a record of original work carried out by us as part of the project work on **MATRICES AND CALCULUS** for B.TECH I Year I Semester (CSE-AIML) students at Malla Reddy Engineering College for Women (Autonomous), Maisammaguda during the academic year 2026-2027.

We further declare that this project work has been carried out by us under the guidance of **(DR.G.Mirona)**, Assistant Professor, Department of Mathematics, and has not been submitted previously to any other institution or university for the award of any degree or diploma.

We also confirm that all the sources of information used in this project have been properly acknowledged and referenced.

Place: Maisammaguda

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Signature of the Students

1) **J SPANDANA**
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CERTIFICATE

This is to certify that the project entitled **"IMAGE COMPRESSION MATRIX RANK REDUCTION"** is a bonafide work carried out **byJ SPANDANA (26RH1A05R9),J VIJETHA (26RH1A05T0) , J SUCHITHRA(26RH1A05T1),** students of B.TECH I Year I Semester (AIML) at Malla Reddy Engineering College for Women (Autonomous), Maisammaguda during the academic year 2025-2026.

The project has been carried out in the Department of Mathematics as part of the academic curriculum for Computer Science and Engineering-AIML students and has been completed under the guidance and supervision of **(DR.G.Mirona).**

This project work is submitted in partial fulfillment of the requirements for the academic evaluation prescribed by Malla Reddy Engineering College for Women (Autonomous), Maisammaguda.

To the best of our knowledge, this work has not been submitted earlier to any other institution or university for the award of any degree or diploma.

Name and Signature of Guide

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Name and Signature of HOD

(Dr. T. Sudha)

Place : Maisammaguda.

Date :

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With Regards and Gratitude

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ABSTRACT

Image compression is an important technique in digital image processing that aims to reduce the amount of storage space and transmission bandwidth required for digital images while preserving the essential visual information. With the rapid growth of digital media, high-resolution images occupy large amounts of storage and require more resources for transmission. Therefore, efficient image compression techniques are necessary for applications such as web services, mobile applications, cloud storage, medical imaging, satellite imaging, and multimedia systems.

The project **"Image Compression Using Matrix Rank Reduction"** focuses on applying concepts of linear algebra to achieve efficient image compression. A digital image can be represented mathematically as a matrix in which each element represents the intensity value of a pixel. The matrix can be decomposed using **Singular Value Decomposition (SVD)** into three matrices: (U), (\Sigma), and (V^T). The singular values in the matrix (\Sigma) represent the importance of different components of the image. By selecting only the most significant singular values and corresponding components, a lower-rank approximation of the original image can be obtained.

In this approach, the original image matrix is replaced by a smaller-rank matrix that contains the major visual information of the image. The compressed image is then reconstructed using the selected singular values. Increasing the rank generally improves image quality but requires more storage, while reducing the rank provides higher compression with some loss of image details. Thus, the method provides a practical trade-off between **compression ratio, storage requirements, and reconstructed image quality**.

The proposed technique demonstrates how mathematical concepts such as **matrices, rank, eigenvalues, and Singular Value Decomposition** can be applied to a real-world problem. The performance of the compression method can be evaluated by comparing the original and reconstructed images using measures such as compression ratio and reconstruction error. The project shows that matrix rank reduction can significantly reduce image data while retaining the important visual characteristics of the original image, making it a useful and educational approach to image compression.

1. INTRODUCTION

Digital images have become an essential part of modern technology and are widely used in communication, education, healthcare, entertainment, social media, satellite systems, and many other applications. As image resolution and quality continue to increase, the amount of data required to store and transmit these images also increases significantly. Large image files require more storage space, higher network bandwidth, and greater transmission time. Therefore, image compression is an important technique for reducing the size of image data while preserving as much useful visual information as possible.

Image compression is the process of representing an image using fewer data elements than the original representation. Compression can be broadly classified into **lossless** and **lossy** compression. Lossless compression allows the original image to be reconstructed exactly, whereas lossy compression removes some information that has relatively less impact on visual quality. The proposed project focuses on a mathematical approach to lossy image compression using **matrix rank reduction**.

A digital image can be represented as a matrix, where each element corresponds to the intensity of a pixel. For a grayscale image, a single matrix can represent the complete image, while a color image can be represented using separate matrices for different color channels. Since many image matrices contain redundant or less significant information, it is possible to approximate them using a matrix of lower rank without completely losing their important visual characteristics.

Singular Value Decomposition (SVD) is one of the most useful mathematical techniques for performing matrix rank reduction. SVD decomposes an image matrix into three matrices, represented as:

[
A = U\Sigma V^T
]

Here, (A) represents the original image matrix, (U) and (V^T) contain important structural information, and (\Sigma) contains the singular values that indicate the relative importance of different components. By retaining only the largest (k) singular values and their corresponding components, a lower-rank approximation of the original image can be created.

The value of (k), known as the **rank**, determines the balance between image quality and compression. A larger value of (k) preserves more details and produces an image closer to the original, but requires more data. A smaller value of (k) provides greater compression but may result in loss of fine details. Therefore, selecting an appropriate rank is an important part of the compression process.

LITERATURE REVIEW

Image compression has been an important research area in digital image processing because of the increasing size and resolution of digital images. Researchers have developed various compression techniques to reduce storage requirements and transmission costs while maintaining acceptable image quality. Traditional compression methods include techniques based on transforms, predictive coding, quantization, and matrix decomposition. Among the mathematical approaches, **Singular Value Decomposition (SVD)** has received considerable attention because of its ability to represent a matrix using a smaller number of significant components.

2.1 Image Compression Techniques

Earlier image compression methods mainly focused on reducing redundancy present in digital images. Standard techniques such as **JPEG** use transform-based compression to represent image information efficiently. These methods have demonstrated that removing less significant information can considerably reduce file size. However, the amount of compression and resulting image quality depend on the characteristics of the image and the selected compression parameters.

2.2 Matrix Representation of Images

An image can be represented mathematically as a matrix of pixel values. In a grayscale image, each matrix element represents the intensity of a pixel. This representation allows mathematical operations to be applied directly to image data. Researchers have explored matrix-based methods because image matrices often contain correlations and redundant information that can be represented using fewer significant components.

2.3 Singular Value Decomposition

Singular Value Decomposition (SVD) is a fundamental technique in linear algebra that decomposes a matrix into three matrices. For an image matrix (A), SVD is represented as:

[
A = U\Sigma V^T
]

The diagonal elements of (\Sigma), called singular values, are arranged in descending order of importance. The larger singular values generally contain more significant information about the structure of the image. By retaining only the first (k) singular values, researchers can construct a lower-rank approximation of the original image.

3. OBJECTIVES

The main objective of the project **"Image Compression Using Matrix Rank Reduction"** is to develop and demonstrate an efficient mathematical approach for reducing the size of digital images while preserving their important visual information.

The specific objectives of the project are:

To understand image representation as a matrix by representing the pixel values of a digital image in the form of a numerical matrix.

To study Singular Value Decomposition (SVD) and understand its role in decomposing an image matrix into significant components.

To apply matrix rank reduction by retaining only the most important singular values and removing less significant components.

To reconstruct the compressed image using a selected number of singular values and compare it with the original image.

To reduce storage requirements by representing the image using fewer matrix components without losing major visual information.

To analyze the effect of different rank values on the quality and size of the reconstructed image.

To measure compression performance using parameters such as compression ratio, Mean Squared Error (MSE), and Peak Signal-to-Noise Ratio (PSNR).

To study the trade-off between compression and image quality, identifying suitable rank values for obtaining an acceptable reconstructed image.

To demonstrate the practical application of linear algebra concepts such as matrices, rank, and SVD in digital image processing.

To develop a simple and understandable image compression system that can demonstrate the effectiveness of low-rank matrix approximation for image data reduction.

METHODOLOGY

The proposed **Image Compression Using Matrix Rank Reduction** system uses mathematical matrix operations and Singular Value Decomposition (SVD) to reduce the amount of data required to represent a digital image. The methodology consists of several sequential stages, starting from image input and ending with the evaluation of the reconstructed image.

4.1 Image Input

The process begins by selecting a digital image from the user's system. The image can be a grayscale or color image. For a grayscale image, the pixel intensity values are represented directly as a two-dimensional matrix. For a color image, the image is separated into individual color channels and each channel is processed independently.

4.2 Image Conversion to Matrix

The selected image is converted into a numerical matrix. Each element of the matrix represents the intensity value of a corresponding pixel. For example, a grayscale image can be represented as:

$$
A =
\begin{bmatrix}
a_{11} & a_{12} & \cdots & a_{1n}\\
a_{21} & a_{22} & \cdots & a_{2n}\\
\vdots & \vdots & \ddots & \vdots\\
a_{m1} & a_{m2} & \cdots & a_{mn}
\end{bmatrix}
$$

This matrix forms the input for the compression process.

4.3 Singular Value Decomposition

The image matrix is decomposed using **Singular Value Decomposition (SVD)**. The decomposition is represented as:

$$
A = U\Sigma V^T
$$

where (U) and (V^T) contain information about the structure of the image, while (\Sigma) contains the singular values. The singular values are arranged in descending order, with the largest values generally representing the most significant information.

TOOLS AND TECHNIQUES

The proposed **Image Compression Using Matrix Rank Reduction** system uses mathematical concepts, programming tools, and image-processing techniques to compress and reconstruct digital images. The following tools and techniques are used in the project.

5.1 Programming Language – Python

Python is used as the primary programming language because it provides simple syntax and powerful libraries for numerical computation, matrix operations, and image processing. It allows the complete compression process to be implemented efficiently.

5.2 NumPy

NumPy is used for numerical computations and matrix operations. It provides functions for creating and manipulating arrays and matrices. The SVD operation can be performed using NumPy's linear algebra functions.

5.3 Image Processing Library

An image-processing library such as **Pillow (PIL)** can be used to open, resize, convert, and save digital images. It helps in converting images into numerical arrays before applying matrix operations and converting the reconstructed matrix back into an image.

5.4 Singular Value Decomposition (SVD)

SVD is the main mathematical technique used in the project. It decomposes an image matrix (A) into three matrices:

[
A = U\Sigma V^T
]

The most significant singular values are retained while less significant values are removed to create a lower-rank approximation.

5.5 Matrix Rank Reduction

Matrix rank reduction is used to reduce the number of components required to represent an image. By selecting a smaller rank (k), the image can be represented using fewer data elements while retaining its major visual features.

IMPLEMENTATION

The **Image Compression Using Matrix Rank Reduction** system is implemented using Python and mathematical matrix operations. The implementation focuses on converting an image into a matrix, applying Singular Value Decomposition (SVD), reducing the matrix rank, reconstructing the image, and evaluating the compression performance.

6.1 Image Selection

The implementation begins by selecting an input image from the computer. The image is loaded using the Pillow library. The image can be converted into grayscale format to simplify the matrix operations. In grayscale images, each pixel is represented by an intensity value ranging from 0 to 255.

6.2 Conversion into Matrix

After loading the image, it is converted into a NumPy array. This array represents the image as a numerical matrix.

For example, an image of size (m \times n) is represented by a matrix:

[
A_{m\times n}
]

Each element in the matrix represents the intensity of a corresponding pixel.

6.3 Applying SVD

The Singular Value Decomposition technique is applied to the image matrix using NumPy. The matrix is decomposed as:

[
A = U\Sigma V^T
]

The matrices (U), (\Sigma), and (V^T) contain the information required to reconstruct the original image.

6.4 Selecting the Reduced Rank

A rank value (k) is selected according to the desired compression level. Only the first (k) singular values and their corresponding components are retained.

For example, if the original matrix has a high rank, selecting a smaller value of (k) significantly reduces the number of components used to represent the image.

RESULTS AND OUTPUT

The **Image Compression Using Matrix Rank Reduction** project was successfully implemented using Python, NumPy, Pillow, and Singular Value Decomposition (SVD). The system takes a digital image as input, converts it into a matrix of pixel values, performs SVD, reduces the matrix rank, and reconstructs the image using only the selected significant singular values.

7.1 Input Image

The original image is first loaded into the system and converted into a grayscale pixel matrix. The matrix contains the intensity values of all pixels in the image.

Input: Original digital image
Representation: Pixel intensity matrix
Processing: Singular Value Decomposition

7.2 SVD Decomposition

After converting the image into a matrix, SVD is applied:

[
A = U\Sigma V^T
]

The singular values obtained from the decomposition are arranged in descending order. The larger singular values contain the most significant information about the image.

7.3 Rank Reduction

Different rank values are selected to generate different levels of compression. For example, ranks such as **10, 25, 50, and 100** can be tested.

A lower rank results in higher compression but may remove more image details. A higher rank preserves

SUMMARY AND DISCUSSION

The **Image Compression Using Matrix Rank Reduction** project demonstrates how mathematical concepts from linear algebra can be applied to solve a practical problem in digital image processing. The main objective of the project was to reduce the amount of data required to represent an image while preserving its important visual information.

In this project, the input image was represented as a matrix of pixel values. **Singular Value Decomposition (SVD)** was then applied to decompose the image matrix into its important components. The singular values were arranged according to their significance, and only a selected number of the largest singular values were retained. This process produced a lower-rank approximation of the original image.

The reconstructed image was obtained using the selected components. The results showed that the rank value has a direct effect on both compression and image quality. When a smaller rank is used, the amount of data required is reduced significantly, resulting in higher compression. However, some fine details of the image may be lost. On the other hand, using a larger rank preserves more image details but provides comparatively less compression.

The project therefore demonstrates an important **trade-off between compression ratio and image quality**. Selecting an appropriate rank is essential for achieving a suitable balance between storage efficiency and visual quality. Performance can be further analyzed using parameters such as **Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and compression ratio**.

8.1 Key Observations

SVD effectively separates important and less significant information in an image matrix.

Matrix rank reduction reduces the number of components required for image representation.

Lower rank values provide greater compression but may reduce image quality.

Higher rank values provide better reconstruction quality but require more data.

The appropriate rank depends on the required balance between storage efficiency and image quality.

APPLICATIONS

Image Compression Using Matrix Rank Reduction has applications in various areas where reducing image data and storage requirements is important. By retaining the most significant information and removing less important components, the technique can be useful in the following areas:

9.1 Digital Image Storage

The technique can reduce the amount of data required to store digital images, helping save storage space on computers, mobile devices, and cloud platforms.

9.2 Web and Social Media

Compressed images require less data for uploading and downloading. This can help improve webpage loading speed and reduce data usage.

9.3 Mobile Applications

Mobile applications frequently handle large numbers of images. Matrix-based compression can help reduce storage requirements and improve image transfer efficiency.

9.4 Cloud Storage

Cloud platforms store huge collections of images. Reducing the representation size of images can help optimize storage capacity and data transfer.

9.5 Image Transmission

Compressed images require less bandwidth during transmission. This can be useful when images need to be transferred over networks with limited bandwidth.

9.6 Medical Imaging

Medical systems handle large amounts of image data such as scans and diagnostic images. Compression techniques can help manage storage and transmission requirements while preserving important visual information.

9.7 Satellite and Remote-Sensing Images

Satellite systems generate large quantities of high-resolution images. Compression can reduce the amount of data that needs to be stored and transmitted from remote locations.

9.8 Digital Photography

Digital cameras and photography applications generate high-resolution images. Compression techniques can help reduce file size while maintaining acceptable image quality.

ADVANTAGES AND LIMITATIONS

10.1 Advantages

Reduces Storage Requirements: Matrix rank reduction decreases the amount of data needed to represent an image, thereby saving storage space.

Maintains Important Image Information: The most significant singular values are retained, allowing the major visual features of the original image to be preserved.

Adjustable Compression: The compression level can be controlled by selecting an appropriate rank value (k).

Simple Mathematical Approach: The technique is based on well-established linear algebra concepts such as matrices, rank, and Singular Value Decomposition.

Easy to Implement: Python libraries such as NumPy and Pillow provide functions that make the implementation relatively simple.

Useful for Learning: The project provides a practical application of linear algebra, numerical methods, and image processing concepts.

Flexible Quality Control: Different rank values can be tested to obtain a suitable balance between image quality and compression.

Reduced Data Transmission: Smaller image representations can require less bandwidth when images are transmitted over a network.

10.2 Limitations

Loss of Image Details: Since smaller singular values are discarded, some fine image details may be lost during compression.

Quality Depends on Rank: Selecting a very low rank can produce noticeable distortion, while a high rank provides less compression.

Computational Cost: Performing SVD on very large, high-resolution image matrices can require considerable computational time and memory.

Not Always Optimal: SVD-based compression may not provide the same practical compression efficiency as specialized image compression standards such as JPEG for every type of image.

Color Image Processing: Color images require processing multiple channels, which

CONCLUSION

The **Image Compression Using Matrix Rank Reduction** project successfully demonstrates how mathematical concepts from linear algebra can be applied to the practical problem of digital image compression. The project uses **Singular Value Decomposition (SVD)** to decompose an image matrix into significant components and reduces its rank by retaining only the most important singular values.

The implementation shows that a digital image can be represented using fewer matrix components while preserving its major visual characteristics. By selecting an appropriate rank value, it is possible to achieve a useful balance between **compression ratio, storage requirements, and reconstructed image quality**. Lower rank values provide greater compression but may result in the loss of some fine details, whereas higher rank values provide better image quality with comparatively less compression.

The project also demonstrates the practical importance of concepts such as **matrix representation, rank reduction, SVD, low-rank approximation, MSE, PSNR, and compression ratio**. The experimental observations confirm that matrix rank reduction can effectively reduce image data while maintaining significant visual information.

Although the method has limitations, particularly computational cost for large images and possible loss of fine details, it provides a simple and effective approach for understanding the principles of image compression. Overall, this project successfully connects theoretical linear algebra concepts with real-world digital image processing and provides a foundation for further research and improvement in efficient image compression techniques.

FUTURE SCOPE

The **Image Compression Using Matrix Rank Reduction** project can be further improved by incorporating advanced techniques and optimizing the existing implementation. The following developments can be considered in the future:

Color Image Compression: Extend the system to efficiently compress RGB and other color images by processing multiple color channels.

Automatic Rank Selection: Develop an algorithm that automatically selects the optimal rank based on image quality and required compression ratio.

Advanced Matrix Decomposition: Explore other decomposition techniques such as **Non-Negative Matrix Factorization (NMF)** and other low-rank approximation methods.

Deep Learning Integration: Combine matrix-based compression with deep learning and neural-network-based image compression techniques for improved performance.

Real-Time Compression: Optimize the algorithm to support real-time image compression for cameras, mobile applications, and streaming systems.

Large Image Optimization: Improve memory management and computational efficiency so that high-resolution images can be processed faster.

Web-Based Application: Develop a user-friendly web application where users can upload an image, select the compression level, and download the reconstructed image.

Batch Image Compression: Add support for processing multiple images automatically instead of compressing one image at a time.

Quality-Based Compression: Allow users to specify the required image quality or PSNR, with the system automatically determining a suitable rank.

Comparison with Standard Methods: Compare SVD-based compression with techniques such as **JPEG, JPEG2000, and other modern compression methods** to evaluate performance under different conditions.

Cloud-Based Implementation: Integrate the system with cloud storage platforms to reduce storage requirements and optimize image transmission.

Improved Evaluation: Include additional image

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